Executive Summary

This project develops a decision support system for optimizing marketing resource allocation using demographic and socio-economic data. The objective is to identify high-value individuals while minimizing unnecessary outreach and missed opportunities.

A classification model is used to estimate income probability, with threshold selection enabling explicit control over the tradeoff between marketing cost and coverage. The model achieves strong discriminative performance and supports efficient targeting decisions under real-world constraints.

In parallel, segmentation identifies distinct population groups based on lifecycle and labor force participation patterns. These segments enable differentiated strategies, including direct targeting of high-value individuals, long-term engagement for growth segments, and deprioritization of low-return groups.

The combined framework integrates prediction and segmentation to improve targeting efficiency and reduce Customer Acquisition Cost (CAC) inefficiency. Governance considerations including weighting, bias monitoring, and model stability controls ensure robustness in real-world deployment.

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Problem Framing & Business Objectives

The core objective is not simply to predict whether an individual earns above 50K, but to support how marketing resources should be allocated under limited budget. In practice, every outreach action has a cost, and not all individuals provide equal expected return. The problem is therefore framed as a resource allocation decision under uncertainty.

The classification task estimates the probability that an individual belongs to a high-income group, which is used as a ranking signal rather than a direct decision. A threshold must be selected to convert probabilities into actions. This threshold represents a business tradeoff. A higher threshold increases precision and reduces unnecessary marketing spend, but it may miss some high value individuals. A lower threshold increases coverage but introduces more low value contacts and higher cost. The optimal choice depends on budget constraints and tolerance for missed opportunities.

The segmentation task complements this by providing structure to the population. Individuals exist within socio-economic contexts and exhibit different patterns of labor participation and financial behavior. Segmentation identifies these groups and enables segment-specific playbooks. Stable high income groups can be prioritized for immediate targeting. Early career groups may require long term engagement. Lower participation groups can be deprioritized due to limited expected return.

By combining classification and segmentation, the system supports two levels of decision making. The model determines who is likely to be high value, while segmentation determines how different groups should be approached. This integrated framework ensures that decisions are both precise at the individual level and strategically aligned at the group level.

Data Understanding & Preprocessing

The dataset is derived from U.S. Census surveys and contains demographic, employment, and financial attributes. Each observation represents an individual, along with a sampling weight that reflects how many people in the population that record represents. This weight is critical and must be incorporated into both model training and evaluation to ensure population-level validity. By using weighted evaluation, we ensure that model performance metrics such as precision and recall reflect population-level impact rather than sample-level statistics.

The target variable is income level, defined as whether an individual earns above 50K. The distribution is highly imbalanced (shown in Figure 1), with a significantly smaller proportion of high-income individuals. This imbalance has direct implications for resource allocation

decisions, as accuracy alone would be misleading. It motivates the use of precision-recall based evaluation and threshold selection aligned with business objectives.

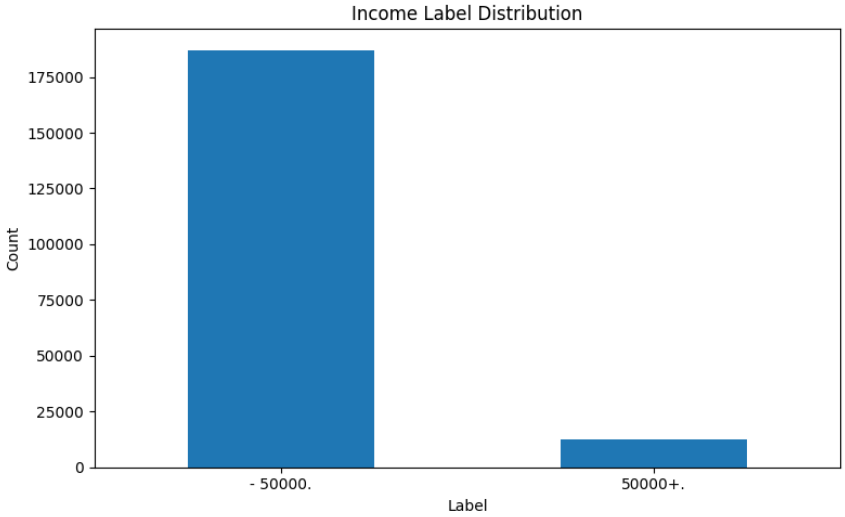

Figure 1: The dataset is highly imbalanced, with significantly fewer high-income observations, motivating the use of precision-recall based evaluation and threshold tuning.

The dataset includes a wide range of categorical variables that capture socio-economic context. Several variables contain values such as “Not in universe” (Figure 2). These values do not represent missing data. They indicate structural absence, such as individuals not participating in the labor force. Preserving these categories is essential to maintain the integrity of the underlying population structure and avoid distortion of the data generating process.

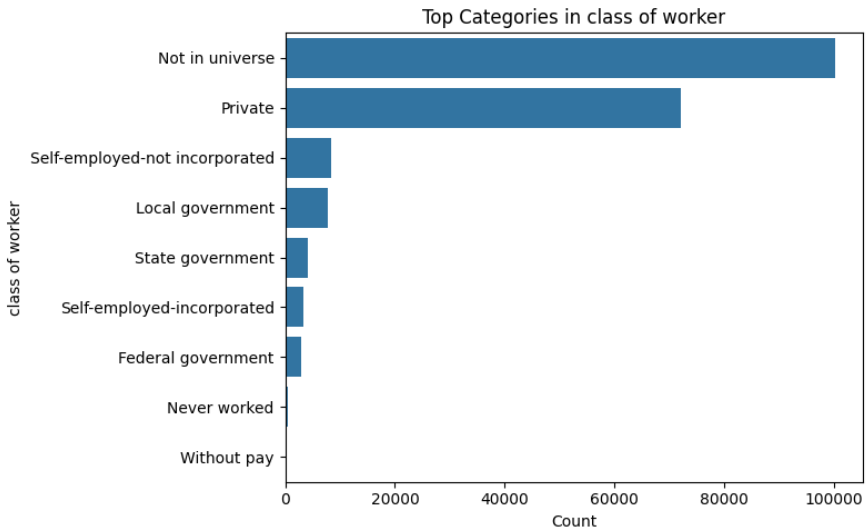

Figure 2: The "Not in universe" category represents structural absence rather than missing data, reflecting individuals not participating in the labor force.

Numeric variables exhibit non-trivial distributional characteristics. Features such as capital gains and financial variables are highly skewed, with large concentrations near zero and a small number of extreme values. To maintain robustness, numeric variables are imputed using median values rather than means. This reduces sensitivity to outliers and aligns with the observed distribution patterns.

Sampling weights are also highly skewed (Figure 3), indicating that observations contribute unevenly to population representation. Ignoring these weights would bias both model training and downstream insights. Therefore, weights are consistently applied across all stages of the pipeline to ensure alignment between model outputs and real-world population effects.

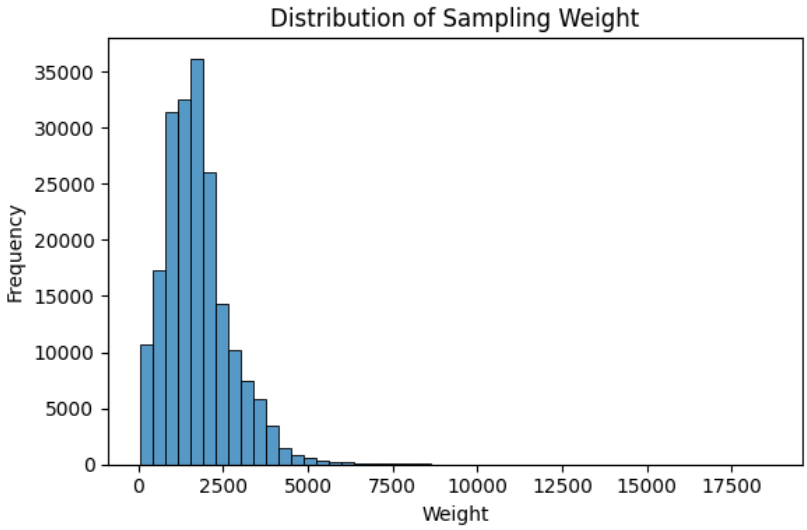

Figure 3: Sampling weights are highly skewed, indicating that observations represent different population scales and must be incorporated into model training and evaluation.

Categorical variables are encoded using one-hot encoding to preserve interpretability and allow the model to capture non-linear relationships. High dimensionality introduced by encoding is managed through model selection and regularization rather than aggressive feature removal.

To prevent data leakage (information contamination), the dataset is split into training, validation, and test sets before any preprocessing steps are applied. All transformations, including imputation and encoding, are fitted only on the training data and then applied to validation and test sets. This ensures that model evaluation reflects true out-of-sample performance and supports pipeline reproducibility.

Overall, the preprocessing strategy preserves the statistical structure of the data while ensuring robustness and consistency. It supports reliable downstream modeling and consistent decision-making under real-world constraints.

Modeling Strategy & Decision Framework

The objective of the classification model is not only to predict whether an individual earns above 50K, but to support resource allocation decisions under business constraints. Each prediction corresponds to a potential marketing action, which introduces a tradeoff between targeting cost and missed opportunity.

Two representative model classes are evaluated: a linear model for interpretability and a tree-based model for capturing non-linear relationships. Logistic Regression and a tree-based model are used to provide sufficient coverage of modeling tradeoffs without introducing unnecessary complexity.

Logistic Regression was selected as the final model due to its interpretability, stability, and consistent precision-recall performance. In addition, it facilitates straightforward monitoring of feature coefficients, enabling early detection of model drift as socio-economic relationships evolve over time. In a financial context, interpretability is critical for understanding model behavior, ensuring transparency, and supporting downstream decision-making.

Model performance is evaluated using ROC AUC as a measure of ranking ability (Figure 4), along with precision and recall to reflect business tradeoffs (Figure 5). The model achieves strong discrimination, indicating that high-income individuals are effectively ranked above low-income individuals. This ensures that high-value individuals are consistently ranked higher, enabling more efficient prioritization in marketing decisions.

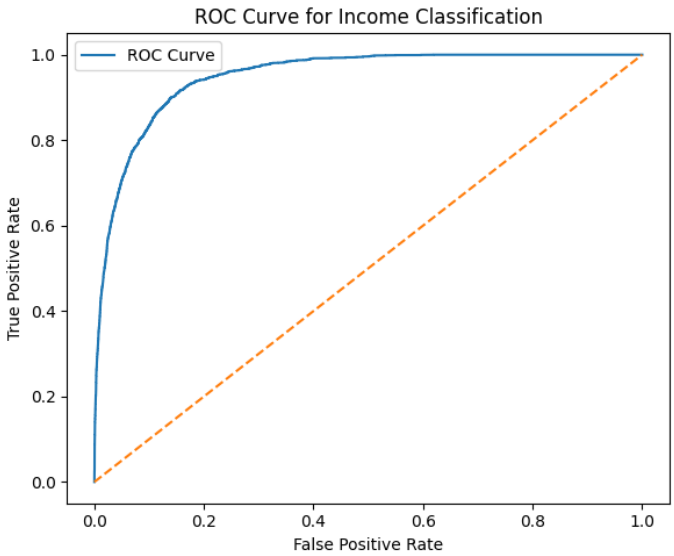

Figure 4: ROC curve demonstrating strong discriminative ability, with the model effectively separating high-income and low-income individuals across decision thresholds.

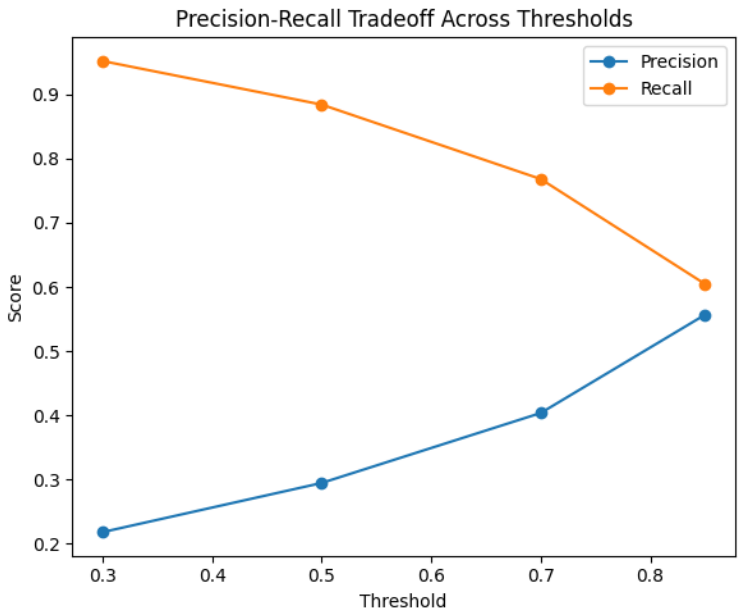

Figure 5: Precision-Recall tradeoff across thresholds. Higher thresholds improve precision and reduce unnecessary outreach, while lower thresholds increase recall at the cost of higher marketing expenditure.

The selected operating point prioritizes precision to maximize return on marketing spend. This reflects a cost-sensitive decision policy aligned with budget-constrained marketing environments. However, model outputs are not directly used for decision-making. Instead, predicted probabilities are converted into actions through threshold selection. The default threshold of 0.5 is not optimal in this context. A higher threshold increases precision, reducing unnecessary marketing costs by targeting fewer but higher-probability individuals. A lower threshold increases recall, capturing more high-income individuals but at the expense of increased outreach cost.

This tradeoff is explicitly analyzed through threshold sensitivity testing. Multiple thresholds are evaluated to understand how precision and recall change under different decision policies. The selected threshold reflects a balance between maximizing return on investment and controlling operational cost.

From a business perspective, this approach transforms the model from a prediction tool into a decision system. Rather than asking whether the model is accurate, the focus shifts to how model outputs are used to optimize resource allocation.

Overall, the modeling strategy prioritizes interpretability, robustness, and alignment with business objectives. It ensures that predictions are actionable and interpretable, supporting reliable decision-making under real-world operational constraints.

Decision Policy

Under budget-constrained conditions, we recommend a precision-prioritized threshold to maximize return on marketing spend.
If budget expands, the threshold can be lowered to increase coverage while accepting higher acquisition cost.
This enables dynamic adjustment of targeting strategy based on business constraints.

Population Segmentation & Actionable Personas

The objective of segmentation is to complement the classification model by structuring the population into meaningful groups. While the model identifies high-income probability, segmentation enables differentiated strategies based on socio-economic context.

Clustering reveals distinct segments reflecting lifecycle stage and labor force participation, capturing broader socio-economic patterns beyond income alone.

Segment Profiles and Business Implications

Segment 1: Prime Earners

Individuals with stable employment, higher education, and strong labor participation. This segment shows the highest income probability. Marketing should prioritize direct targeting of premium products such as high-limit credit cards and wealth management services. A higher threshold ensures efficient allocation of resources while maintaining strong conversion potential.

Segment 2: Career Builders

Early-stage professionals with moderate education and growing participation. While current income is lower, they show upward mobility. This segment is best suited for long-term engagement through entry-level financial products, education, and loyalty programs to build future value.

Segment 3: Transitional Workforce

Individuals with irregular or partial employment. Income probability is moderate to low and less stable. Marketing should be selective and cost-conscious, focusing on targeted outreach rather than large-scale campaigns.

Segment 4: Low Participation Segment

Individuals with limited or no labor force participation, often reflected in structural categories such as “Not in universe.” Income probability is consistently low. This segment should be

deprioritized to reduce unnecessary outreach and improve cost efficiency.

Integrated Decision Framework

The combination of classification and segmentation forms a two-layer decision system. The model determines who to target, while segmentation determines how to target them. This enables segment-specific playbooks aligned with predicted value and socio-economic context.

Business Impact

From a resource allocation perspective, segments can be prioritized as follows:

Segment 1 > Segment 2 > Segment 3 > Segment 4 based on expected return and conversion potential.

By integrating model predictions with segmentation, the system improves targeting efficiency and reduces Customer Acquisition Cost (CAC) inefficiency. Resources are concentrated on high-return individuals, while low-probability segments are filtered out. This shifts the focus from prediction accuracy to decision effectiveness, ensuring measurable business impact.

Model Risk & Governance

Aligned with internal Model Risk Management (MRM) frameworks, the model is subject to several limitations that must be considered in real-world deployment. The dataset is derived from historical census data and may not fully reflect current socio-economic conditions, which introduces the risk of performance degradation over time.

Potential bias may arise from demographic variables such as education, occupation, or marital status. These variables may correlate with sensitive attributes and require careful monitoring to ensure fair and responsible use.

To mitigate these risks, ongoing model monitoring is required. We recommend establishing baseline metrics such as Population Stability Index (PSI) and Characteristic Stability Index (CSI) to track distribution shifts in key drivers, including capital gains and occupation, on a periodic basis. In addition, model performance metrics should be continuously evaluated to detect degradation in predictive quality.

Overall, these considerations ensure that the model remains robust, fair, and aligned with governance standards, supporting reliable decision-making in a production environment. These controls ensure that model outputs remain auditable, stable, and aligned with regulatory expectations. These monitoring controls enable the model to be deployed with clear auditability and periodic performance review, supporting safe integration into

production workflows.

Conclusion

This project develops an end-to-end decision support system for optimizing marketing resource allocation using demographic and socio-economic data. Rather than focusing solely on prediction accuracy, the approach integrates classification and segmentation to support practical, business-aligned decision-making.

The classification model provides a robust ranking of individuals by income probability, while threshold selection enables explicit control over the tradeoff between marketing cost and missed opportunity. Logistic Regression is selected for its interpretability, stability, and ease of monitoring, ensuring that model behavior remains transparent and manageable in a production environment.

Segmentation complements the model by introducing structured population insights. By identifying distinct socio-economic groups, the system enables segment-specific playbooks that align outreach strategies with both predicted value and underlying customer characteristics.

Together, these components form an integrated framework that improves targeting efficiency and reduces Customer Acquisition Cost (CAC) inefficiency. High-value individuals are prioritized, growth segments are strategically nurtured, and low-return segments are filtered out, resulting in more effective and scalable marketing operations.

Finally, governance considerations such as model monitoring, drift detection, and bias awareness ensure that the system remains reliable over time. This reinforces the transition from a predictive model to a production-ready decision system that supports sustained business impact and scalable decision-making in real-world environments.